



UNIVERSITY OF SRI JAYEWARDENEPURA

**Trend Analysis and Forecasting of Tourist Arrivals
in Sri Lanka**

STA 351 2.0 Research Methodology

SUBMITTED BY

DATA TEAM

S. PAHALAGE – AS2021388

W.H.S.C. WIJAYARATHNA – AS2021450

I.A.A. WATHSALA – AS2021506

S.P.O.V. PATHIRATHNA – AS2021541

R.T.T. BUDUNWELA – AS2021571

SUBMITTED TO

DR. NELUKA DEVPURA

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Abstract

This report analyzes recent trends in tourist arrivals in Sri Lanka, focusing on data from 1971 to 2023. It examines various factors that influence the number of tourists visiting the country, including the time of year, the purpose of their visit, how long they stay, and their age, gender, and occupation. The report identifies patterns by looking at these factors, such as peak tourist seasons and demographic differences in visitors.

The findings reveal that tourist arrivals in Sri Lanka are influenced by seasonal factors, with certain months showing higher numbers of visitors. Additionally, the report indicates that the reasons for visiting Sri Lanka vary, with some tourists coming for leisure, while others visit for business or other purposes. The duration of stays also differs, and these differences are connected to the tourists' backgrounds, such as their age or occupation.

This report provides useful insights for those involved in Sri Lanka's tourism industry. Understanding these trends can help plan and improve services to meet the needs of different types of tourists. The report concludes with recommendations on attracting more visitors and enhancing their experience, ensuring that Sri Lanka remains a popular destination.

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Trend Analysis and Forecasting of Tourist Arrivals in Sri Lanka

1. Introduction

1.1 Overview of Tourism in Sri Lanka

Tourism is a key part of Sri Lanka's identity. The island is known for its natural beauty, including stunning beaches, lush tea plantations, and wildlife-filled national parks. Sri Lanka also has a rich cultural heritage, with ancient cities, temples, and festivals that attract visitors from around the world. Over the years, Sri Lanka has built a reputation as a top tourist destination, welcoming millions of travelers each year who come to experience its unique attractions.

The tourism industry in Sri Lanka has gone through many changes, especially after the end of the civil conflict in 2009. Before the conflict, tourism was already an important part of the country, but the years of unrest caused a significant decline in the number of visitors. However, with the return of peace, Sri Lanka experienced a rapid increase in tourist arrivals, making tourism one of the fastest-growing sectors in the country. This growth has brought many opportunities, but it also comes with challenges that need careful management to ensure the long-term sustainability of the industry.

The COVID-19 pandemic brought new challenges, causing a sharp drop in tourist arrivals due to travel restrictions and health concerns. The pandemic showed how vulnerable the tourism sector is to global events, leading to a temporary stop in its growth. As the world recovers, Sri Lanka's tourism industry must rebuild and adapt to new travel conditions, focusing on being resilient and sustainable to ensure a strong future.

1.2 Importance of Tourism to Sri Lanka

Tourism in Sri Lanka is important for many reasons. It brings people from different parts of the world together, promoting cultural exchange and understanding. This exchange helps to preserve and share Sri Lanka's rich cultural traditions and natural heritage. Tourism also plays a crucial role in supporting local communities. Many people in Sri Lanka rely on tourism for their livelihoods, whether they work in hotels, run restaurants, operate tours, or sell handicrafts. The income generated from tourism helps improve the quality of life for many families and communities across the country.

Moreover, tourism helps to drive improvements in infrastructure, such as better roads, airports, and public facilities, which benefit both visitors and residents. The industry also encourages the preservation of the environment and cultural sites, as these are key attractions for tourists. By analyzing trends in tourist arrivals, we can better understand what drives people to visit Sri Lanka and how the country can continue to attract more tourists in the future.

1.3 Objectives of the Study

The main purpose of this study is to explore and analyze the recent trends in tourist arrivals to Sri Lanka. We will look at various factors, including when tourists visit, why they come, how long they stay, their age, gender, occupation, and where they come from. By examining these aspects, we aim to identify patterns and trends that can help us understand how tourism in Sri Lanka has evolved. The other purpose of this research is to develop the best model for forecasting monthly tourist arrivals in Sri Lanka.

This study also seeks to provide insights that can help those involved in the tourism industry, such as business owners, policymakers, and tourism planners, to make informed decisions. By understanding the preferences and behaviors of different groups of tourists, the industry can develop strategies to attract more visitors, improve the overall tourist experience, and ensure that tourism continues to be a positive force for Sri Lanka.

1.4 Scope and Limitations of the Study

This report focuses on data from 1971 to 2023, offering a detailed analysis of how tourist arrivals have changed over this period. The study covers a wide range of factors, including the time of year, reasons for visiting, length of stay, and the demographic profile of tourists. However, it is important to note that the study is based on the available data and does not include more recent developments or the impact of global events like the COVID-19 pandemic. Despite these limitations, the findings of this report provide valuable insights into the long-term trends in Sri Lanka's tourism sector and can serve as a foundation for future studies.

2. Literature Review

Tourism is a vital component of Sri Lanka's economy, significantly contributing to employment, foreign exchange earnings, and overall economic growth. Over the years, the industry has evolved through various phases, shaped by a range of internal and external factors, such as political conflicts, natural disasters, and global pandemics. These factors have created both opportunities and challenges for the tourism sector, influencing its development and sustainability. For instance, as noted by Garín-Muñoz (2007), tourism demand is highly sensitive to economic and social conditions, which can either boost or hinder growth depending on the prevailing circumstances.

Sri Lanka's tourism industry has also been impacted by specific regional trends and global economic shifts. The International Recommendations for Tourism Statistics (IRTS, 2008) emphasize the importance of accurate data collection and analysis to understand these trends better and to make informed policy decisions. In this context, forecasting models, such as the Seasonal Auto-Regressive Integrated Moving Average (SARIMA), have become essential tools for predicting tourist arrivals and planning for the future. As highlighted by Box and Jenkins (1970) and later applied in studies like those by Konarasinghe (2016) and Makoni and Chikobvu (2018), these models help in capturing the seasonal patterns and potential disruptions that can affect tourism flow.

This paper aims to provide an in-depth analysis of tourism patterns, the impact of economic factors, the effectiveness of forecasting models, demographic trends, and the challenges faced by Sri Lanka's tourism industry. The study draws on a range of academic sources, including time series analyses, economic impact studies, and demographic research, to offer a comprehensive overview of the current state and future prospects of tourism in Sri Lanka.

2.1 Tourism Patterns and Regional Analysis in Sri Lanka

Tourist arrivals in Sri Lanka have shown distinct patterns across different regions. Konarasinghe (2016) conducted an analysis using monthly data from January 2008 to December 2014, focusing on four major regions: Asia, Western Europe, Eastern Europe, and the Middle East. The study employed Descriptive Statistics, Time Series plots, and Auto-Correlation Functions to identify trends, while One-way Analysis of Variance (ANOVA) was used to compare mean arrivals from these regions. The findings indicated significant differences in tourist arrivals, particularly between Asia and Western Europe compared to other regions. An increasing trend was observed across all regions, with a strong correlation between current and past tourist arrivals. The study concluded that the Seasonal Auto-Regressive Integrated Moving Average (SARIMA) technique is more effective for forecasting tourist arrivals than the standard Auto-Regressive Integrated Moving Average (ARIMA) model.

2.2 Impact of Economic Factors and External Shocks on Tourism

The tourism industry in Sri Lanka has been significantly influenced by various economic factors and external shocks. Suresh and Senthilnathan (2013) explored the causal relationship between economic growth and tourism earnings in Sri Lanka from 1977 to 2012, using the Granger Causality test. Their study revealed a uni-directional causality where economic growth drives tourism earnings. Similarly, Francis and Salahudeen (2022) examined the relationship between tourism earnings and economic growth from 1970 to 2020, considering the impact of the COVID-19 pandemic. They found a long-run equilibrium and a causal relationship between tourist arrivals and earnings, highlighting the importance of government initiatives to support and boost international travel demand. The Easter attacks in 2019 and the COVID-19 pandemic further emphasized the vulnerability of the tourism sector to external shocks, with the pandemic causing a 70.8% decline in international arrivals in March 2020 (SLTDA, 2020).

2.3 Forecasting Tourist Arrivals Using SARIMA Models

Accurate forecasting of tourist arrivals is essential for effective tourism planning and policymaking in Sri Lanka. Thushara, Su, and Bandara (2019) emphasized the importance of using SARIMA models to predict tourist arrivals, assessing forecast accuracy with data from 2017. Their study highlighted the necessity of better infrastructure, scientific forecasting techniques, and strategic marketing campaigns in key regions to meet government targets. Additionally, Makoni and Chikobvu (2018) applied SARIMA models to forecast tourist arrivals in Zimbabwe, finding the model effective in capturing seasonal patterns. These studies underscore the importance of robust forecasting models, especially in the face of external shocks like political instability and natural disasters, to ensure the resilience and sustainability of the tourism industry.

2.4 Demographic Trends and Seasonality in Tourism

Understanding demographic trends and seasonality in tourist arrivals is crucial for the sustainable development of Sri Lanka's tourism industry. Perera (2017) analyzed trends in tourist arrivals from 1986 to 2015, noting significant increases in overall arrivals, with peak periods in January to March and December, and lower periods in May and June. The study also observed demographic shifts, including an increase in older tourists and a decline in younger ones. Additionally, there was a notable rise in female tourists and increased immigration from regions such as South Asia, Eastern Europe, the Middle East, and Australasia. These trends offer valuable insights for tourism stakeholders, aiding in strategic planning and policy development.

2.5 Challenges and Future Directions in Sri Lanka's Tourism Industry

The tourism industry in Sri Lanka has faced numerous challenges over the years, including the civil war, the Easter attacks, and the COVID-19 pandemic. Mustafa (2019) highlighted the impact of tourism direct employment and earnings on tourist arrivals, emphasizing the strong correlation between these factors. The study suggests that improving employment and wages in the tourism industry should be a priority for Sri Lanka's authorities to enhance tourist arrivals. Furthermore, the need for accurate forecasting models, such as SARIMA, and the importance of strategic management and policy execution are crucial for optimizing the economic benefits of tourism. As the industry continues to recover from recent challenges, it is essential to focus on non-economic factors, such as demographic trends and the effects of historical events, to develop a resilient and sustainable tourism sector in Sri Lanka.

3. Methodology

3.1 Data Collection and Sources

This study is based mainly on secondary data. We collected data from the Annual Statistical Reports published by the Sri Lanka Tourism Development Authority (formerly known as the “Sri Lanka Tourist Board”). We gathered monthly tourist arrival data for 52 years from 1971 to 2023. Additionally, we collected data on the annual percentage distribution of tourist arrivals for 30 years from 1991 to 2020, focusing on key characteristics of tourists.

The study looks at five main characteristics: the seasonality of arrivals, purpose of visit, length of stay, age and gender distributions, and occupational categories. To understand the trends and seasonality, we analyzed monthly tourist arrivals over 50 years. We also collected data on annual tourist arrivals by different regions for the 30 years from 1993 to 2023.

3.2 Data Analysis

The data analysis for this study was conducted using both Minitab and R software. Descriptive statistics, which summarized the socio-demographic characteristics of the tourists—such as age, gender, occupation, and purpose of visit, as well as the seasonal trends in tourist arrivals—were carried out using Minitab. Regression analysis and trend analysis were also performed using Minitab to explore relationships between different variables, such as the purpose of the visit and length of stay. Time series analysis was conducted using R software, and both Minitab and R were employed to fit the most suitable SARIMA model for forecasting tourist arrivals. The significance level (α) for all statistical tests was set at 0.05, ensuring the reliability of the findings. Additionally, outliers in the data were carefully reviewed to maintain the accuracy of the analysis.

3.3 Limitations

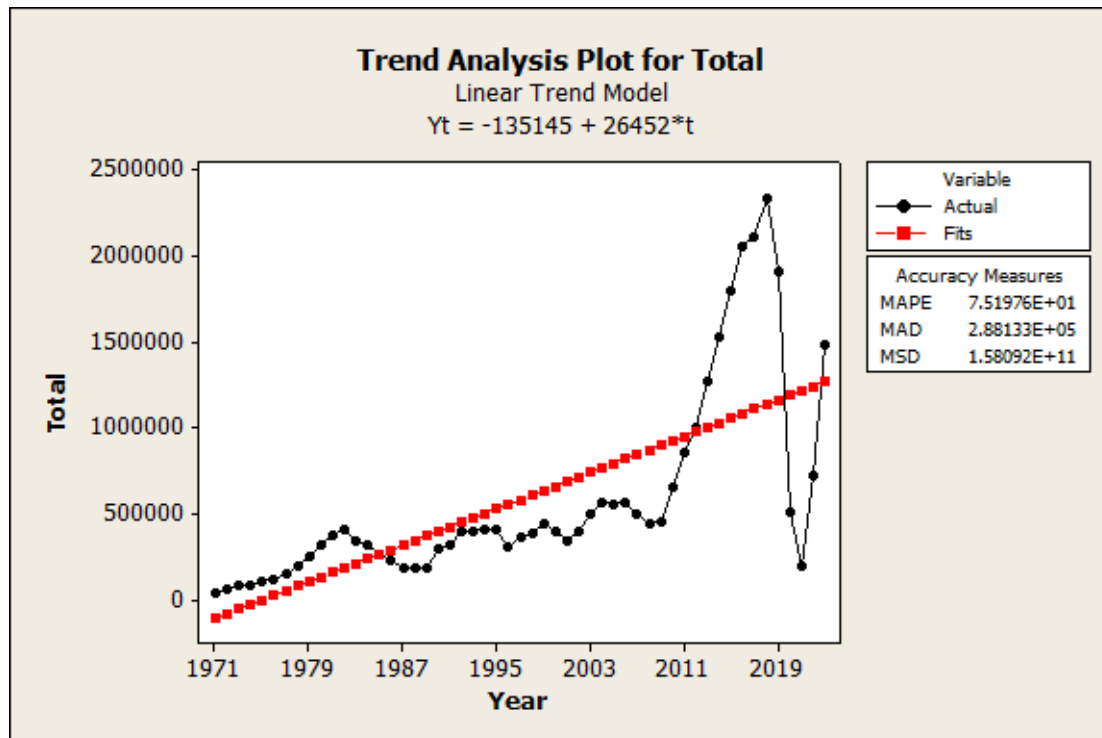
This study used secondary data from the Sri Lanka Tourism Development Authority, covering the period from 1971 to 2023. While this data is valuable, it may have some limitations in detail and scope, and it might not fully capture recent linear trends or new factors. The analysis focuses on key tourist characteristics like seasonality, purpose of visit, duration of stay, age, gender, and occupation. This may limit the understanding of other influences on tourist arrivals, such as changes in global travel behavior or economic conditions

4. Result and Finding

Overall Trends in the Number of Tourist Arrivals

The analysis of annual tourist arrivals in Sri Lanka from 1971 to 2023 shows a moderate increasing trend. Linear regression reveals that the number of tourist arrivals increases by approximately 26,452 each year. The R^2 value of 0.509 indicates that the model explains about 50.9% of the variability in the number of tourist arrivals. This suggests a moderate correlation between the years and the increase in tourist numbers. Overall, Sri Lanka has seen a steady growth in international tourist arrivals over the past three decades.

Figure 1- Trend Analysis Plot for Total



Model Summary:

S	R-sq	R-sq(adj)	R-sq(pred)
405329	50.88%	49.91%	45.60%

Coefficients:

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	-135145	112947	-1.20	0.237	
t	26452	3640	7.27	0.000	1.00

Seasonal Patterns of Tourist Arrivals – 1971 to 2023

In statistical analysis, a 12-month moving average method was utilized to investigate the trends in the seasonal variations of tourist arrivals from 1971 to 2023. Monthly data on the total number of tourist arrivals was sourced from the Annual Statistical Reports, published by the SLTDA, for the analysis. The outcomes of the Seasonal Indices of tourist arrivals for 12 months are presented in Table 1.

Table 1: *Seasonal Indices of tourist arrivals of 12 months*

Month	Seasonal Indices of tourist arrivals
January	1.26275
February	1.22553
March	1.19746
April	0.90102
May	0.69053
June	0.70613
July	1.00147
August	0.99559
September	0.80866
October	0.87534
November	1.02908
December	1.30645

Figure 2 - *Seasonality of Tourist Arrivals 1971 to 2023*



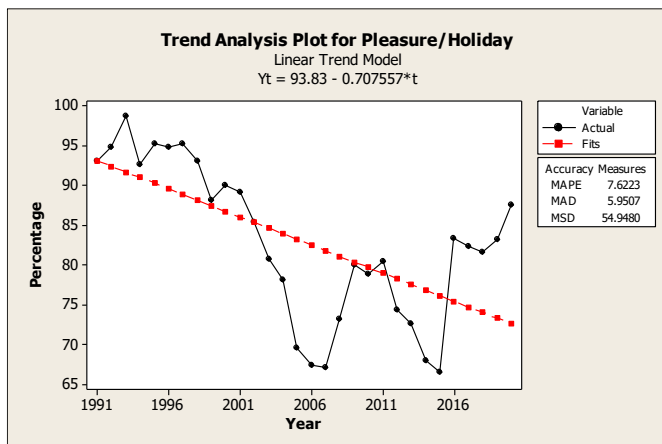
Figure 2 illustrates the seasonal variations in tourist arrivals from 1971 to 2023. Tourist arrivals are highest during January and December, with the seasonal index peaking at around 1.3 in December and slightly above 1.25 in January, indicating these months attract more tourists than the average. From February to June, the seasonal index declines, reaching its lowest point in May and June, with values around 0.7, suggesting a significant drop in tourist numbers. In July and August, the index rises back to around 1.0, before decreasing again in September. The trend then shifts upward in October and November, culminating in the December peak. So, the lowest number of arrivals was in May and June and the highest number of tourists arrived in January, February, and December.

Purpose of visit

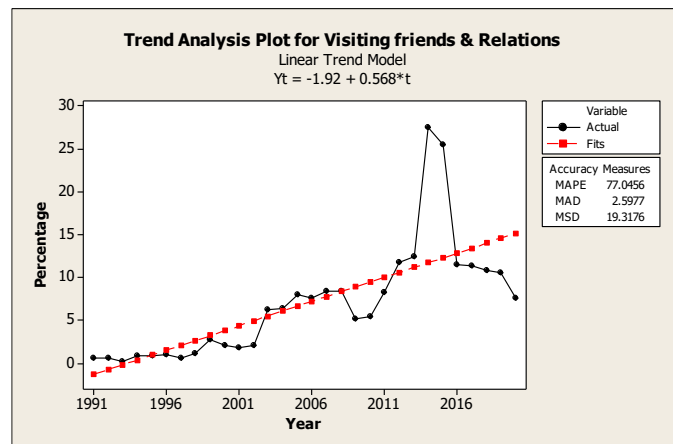
To investigate the trends in the percentage distribution of tourists by their purpose of visit from 1991 to 2020, annual data on the percentage distribution of tourists was sourced from the Annual Statistical Reports published by the SLTDA. As the (SLTDA, 2013), the primary reasons for traveling to Sri Lanka included pleasure/holiday, visiting friends and relatives, religious and cultural activities, business, and other purposes. The Department of Immigration and Emigration expanded the categories to include health, sports, education, and official purposes (SLTDA, 2014). Due to the limited data points in these new categories, the analysis was conducted using the old categories. Linear Regression analysis was performed. The graphical representation clearly shown in Figure 3 and Table 2 provides a summary of the regression analysis to visit Sri Lanka.

Figure 3: Trends in Percentage Distribution of Tourists by Purpose of Visit - 1991 to 2020

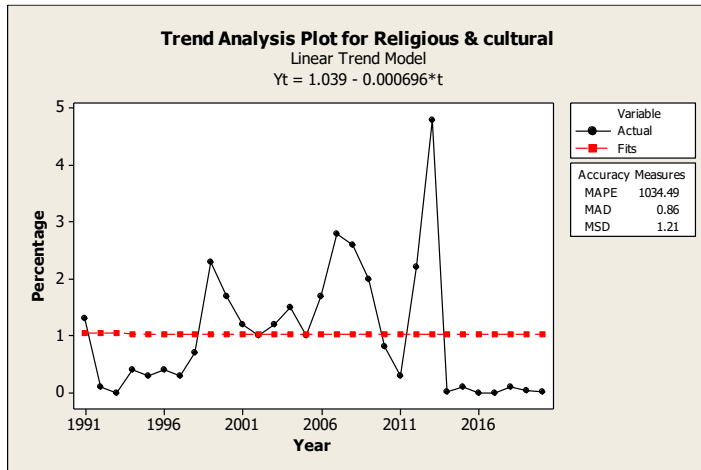
Pleasure/Holiday:



Visiting friends & Relations:



Religious & cultural:



Business:

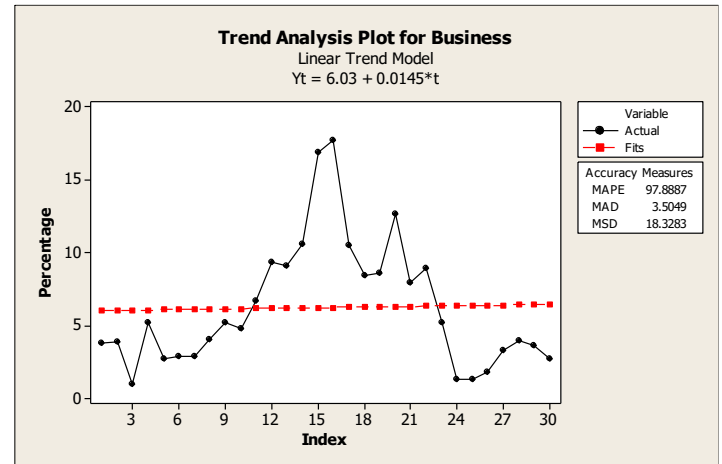


Figure 3 illustrates the changes in the percentage of tourists by their purpose of visit from 1991 to 2020. This graph highlights that the reasons for visiting vary widely. It shows a small increase in tourists visiting for business and a small decrease in those visiting for religious and cultural reasons between 1991 and 2020. Additionally, the trends for tourists visiting for pleasure or holiday are dropping quickly, while those visiting friends and family are rising sharply.

Table 2: Regression Parameters for Annual Percentage Distribution of Tourists by Purpose of Visit

Purpose of Visit	m (Slope)	c (Intercept)	R square (R2)	Trend	Strength of relationship
Pleasure/Holiday	-0.7076	93.833	0.406	Decrease	Moderate
Business	0.01451	6.029	0.001	Increase	Very weak
Visiting Friends & Relations	0.5677	-1.918	0.556	Increase	Moderate
Religious & cultural	-0.00069	1.0391	0.0	Decrease	Very weak

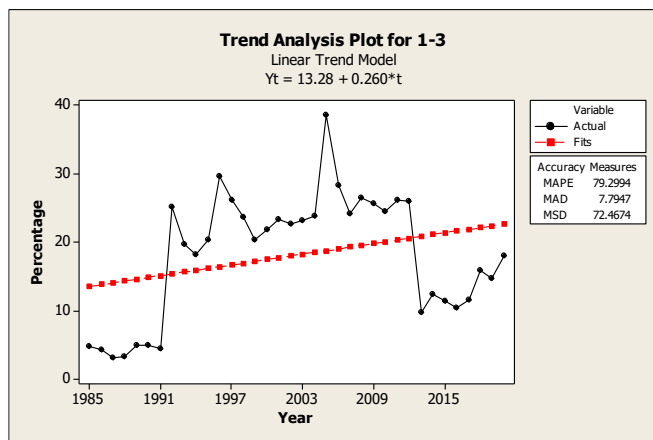
The table 2 presents the results of a linear regression analysis on the annual percentage distribution of tourists by their purpose of visit to Sri Lanka. For tourists visiting for pleasure or holiday, the negative slope (-0.7076) indicates a decreasing trend over time, with a moderate model fit with R^2 value 0.406. Business travel shows a very weak increasing trend with a positive slope (0.01451) and a very low R^2 value of 0.001, indicating a very weak relationship. Visits to friends and relatives exhibit an increasing trend, as shown by the positive slope (0.5677) and a moderate model fit with R^2 value of 0.556. Lastly, religious, and cultural visits show a decreasing trend with a near-zero slope (-0.00069) and an R^2 value of 0.0, indicating a very weak relationship.

Duration of stay

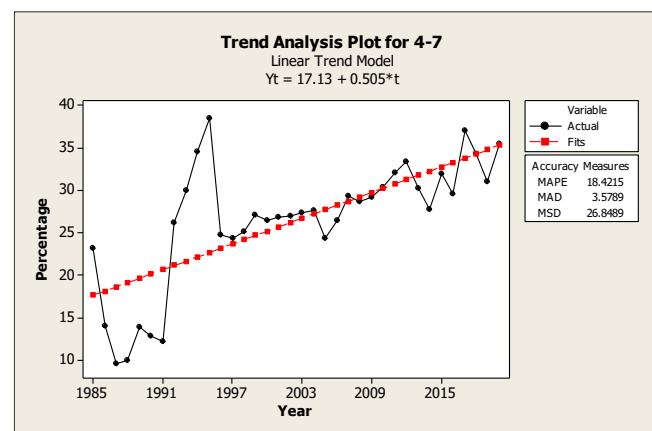
The number of tourists visiting and the duration of their stay in the country are both important. Therefore, statistical analysis was conducted using 36 years of tourist records categorized by length of stay (number of nights). Linear regression analysis was performed to identify trends in the annual percentage for each length of stay between 1985 and 2020. Figure 4 graphically shows the linear trends in the percentage distribution of tourists by length of stay for the period from 1985 to 2020. Table 3 summarizes the results of the regression analysis for the length of stay.

Figure 4: Trends in Percentage Distribution of Tourists by Period of Stay – 1985 to 2020

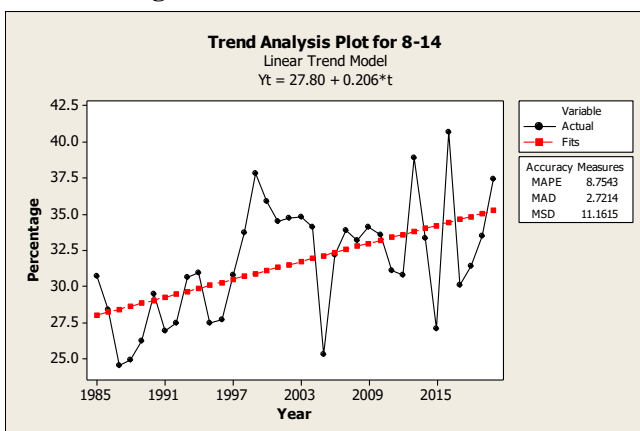
1-3 Nights



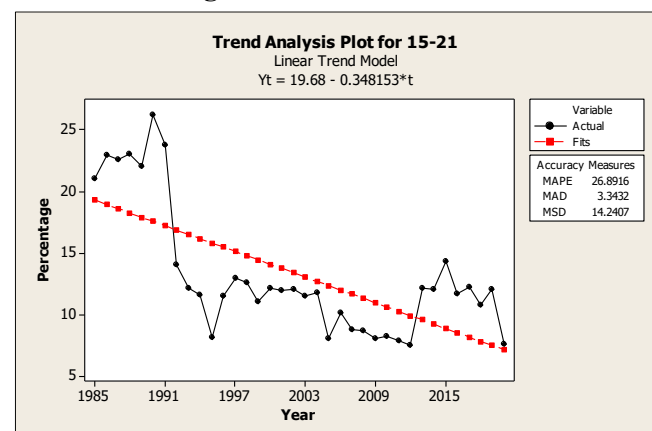
4-7 Nights



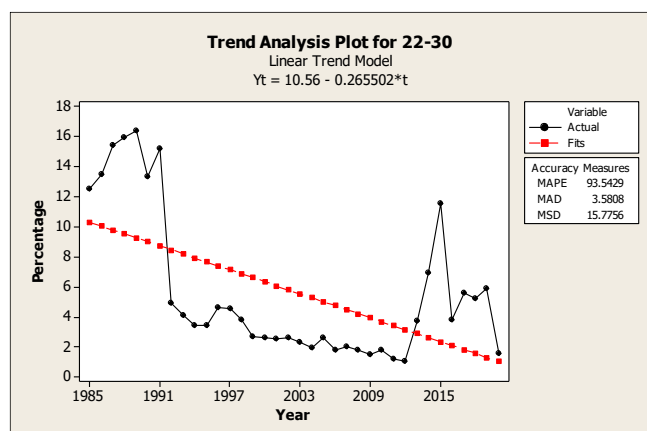
8-14 Nights



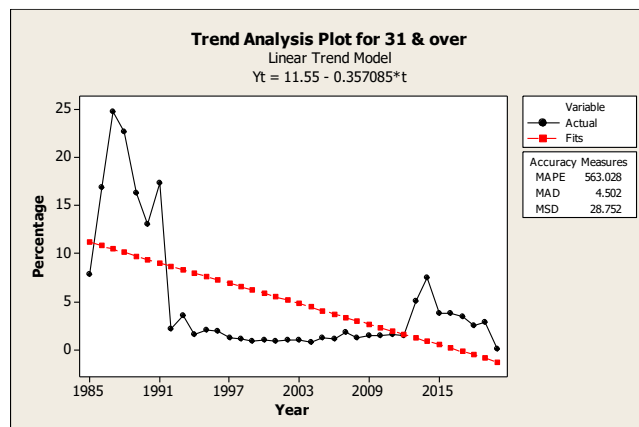
15-21 Nights



22-30 Nights:



31 Nights & over:



Regarding the graphical illustration, the 1-3 nights category shows a gradual linear increase from 1985 to 2020. However, there is a significant increase from 1991 to 1992 and a notable decrease from 2012 to 2013. Both the 4-7 nights and 8-14 nights categories exhibit a rapid increase from 1985 to 2020. In contrast, the 15-21 nights, 22-30 nights, and 31 & over categories show a rapid decrease over the same period.

Table 3: Regression Parameters for Annual Percentage Distribution of Tourists by Period of Stay

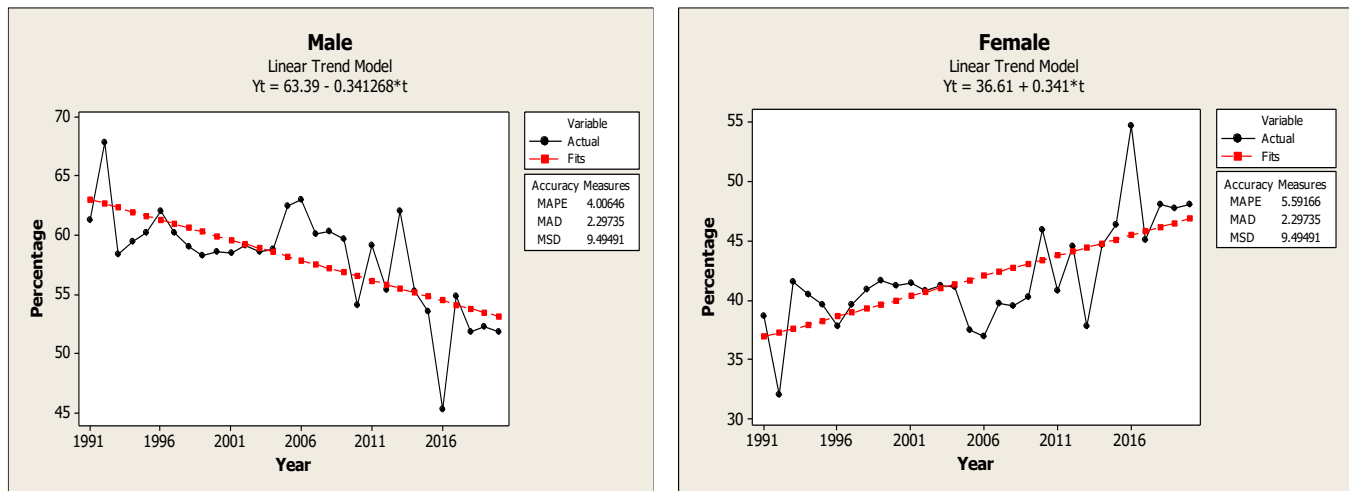
Period of stay (nights)	m (Slope)	c (Intercept)	R square (R ²)	Trend	Strength of relationship
1-3	0.260	13.28	0.091	Increase	Very weak
4-7	0.505	17.13	0.506	Increase	Moderate
8-14	0.206	27.8	0.292	Increase	weak
15-21	- 0.348	19.68	0.479	decrease	Moderate
22-30	- 0.266	10.56	0.325	decrease	weak
31 & over	- 0.357	11.55	0.324	decrease	weak

Table 3 provides a summary of the regression analysis for the annual percentage distribution of tourists by their period of stay in Sri Lanka, emphasizing the strength of the linear trend model for each category. For tourists staying 1-3 nights, the slope is 0.260, indicating an increasing trend. However, the R-squared value is 0.091, suggesting a very weak model fit, which means the linear trend model does not explain much of the variability in the data. Tourists staying 4-7 nights also show an increasing trend with a slope of 0.505 and a moderate R-squared value of 0.506, indicating a moderately strong linear trend model.

The 8-14 nights category has a weak increasing trend with a slope of 0.206 and an R-squared value of 0.292, reflecting a weak model fit. For the 15-21 nights category, the slope is -0.348, indicating a decreasing trend, and the R-squared value is 0.479, suggesting a moderately strong model fit. Tourists staying 22-30 nights show a decreasing trend with a slope of -0.266 and a weak R-squared value of 0.325, implying a weak linear trend model. Lastly, for stays of 31 nights and over, the slope is -0.357, indicating a decreasing trend, with an R-squared value of 0.324, also suggesting a weak model fit.

Gender and Age

Figure 5: Trends in Percentage Distribution of Tourists by Gender - 1991 to 2020



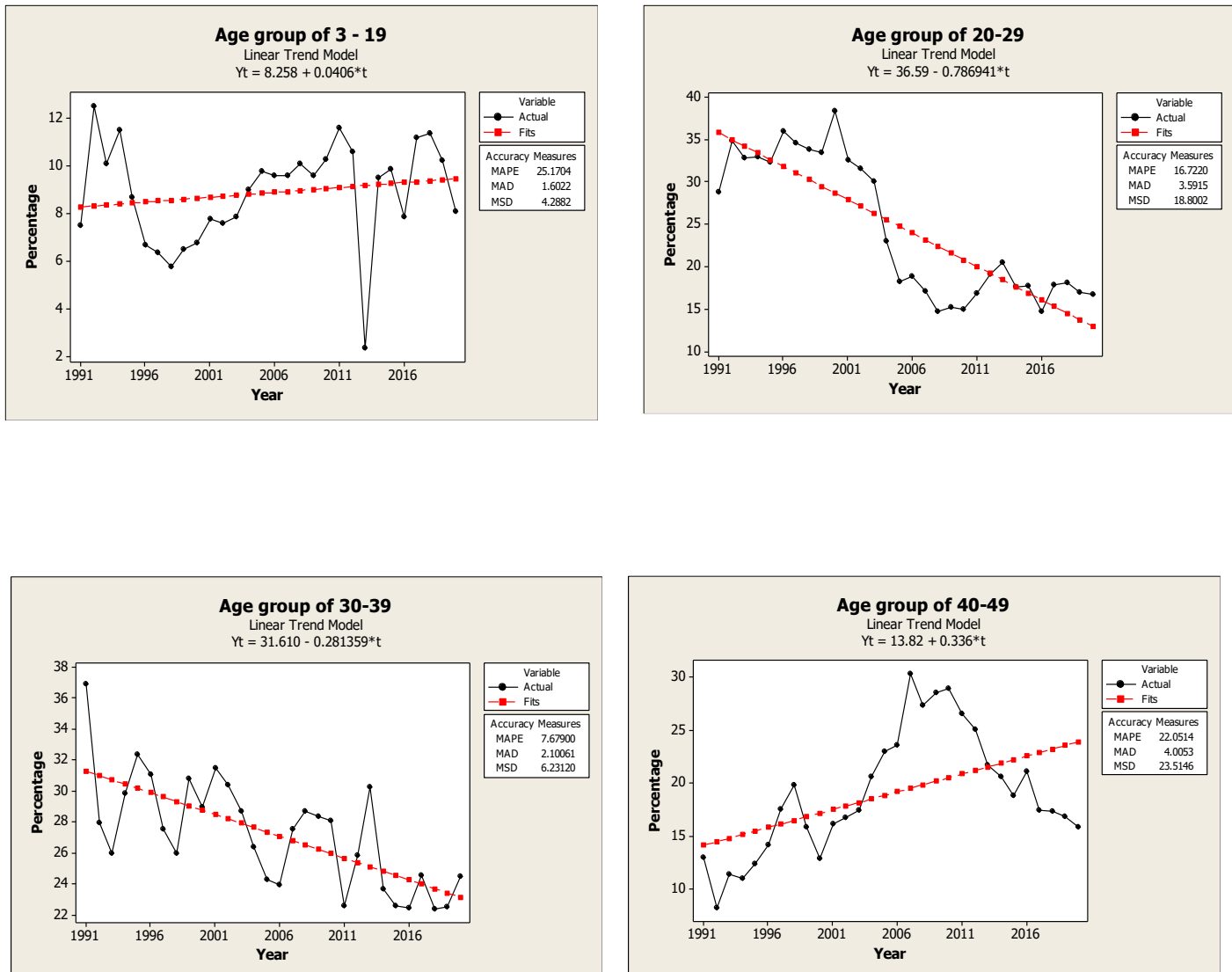
The analysis focused on the trends in tourist arrivals in Sri Lanka, specifically examining the changes in the percentage of male and female tourists from 1991 to 2020. Figure 5 shows the percentage distribution of tourist's gender. According to the table 4, it is found that the percentage distribution of male representation has been decreasing moderately, while the trend in female representation has been increasing moderately. Both trends have a moderate relationship with time between 1991 and 2020, as indicated by the R^2 values of 0.479.

Table 4: Regression Parameters for Annual Percentage Distribution of Tourists by Gender

Gender	m (Slope)	C (Intercept)	R-square (R^2)	Trend	Strength of relationship
Male	- 0.341	63.4	0.479	Decrease	Moderate
Female	0.341	36.6	0.479	Increase	Moderate

Figure 6 represent the percentage distribution of tourists by Age Groups between 1991 to 2020. When considering the age group of tourists, the annual percentage of tourists has decreased in the age group between 20-29 years and between 30 to 39 years. The highest rate of increasing trend in annual percentage is shown in the age group between 50-59 years, with a rate of 0.733 percent per year. While the lowest rate of increasing trend is represented in the age group between 3 to 19, with a rate of 0.028 percent. According to the overall result, the younger tourists have demonstrated a decreasing trend and the older tourists have shown an increasing trend to visit Sri Lanka from 1991 to 2020.

Figure 6: Trends in Percentage Distribution of Tourists by Age Groups – 1991 to 2020



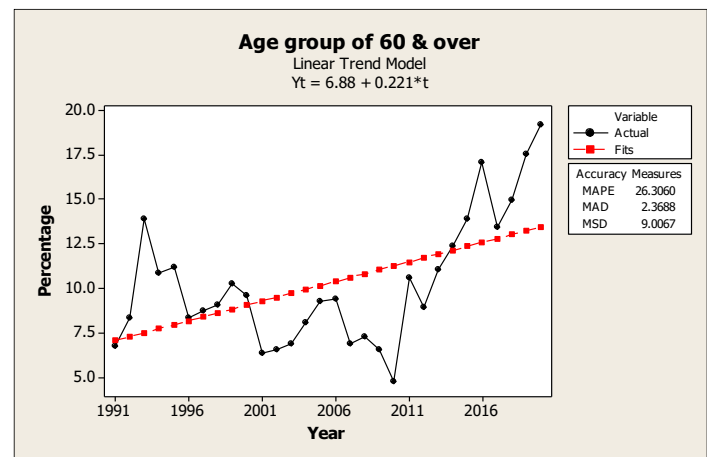
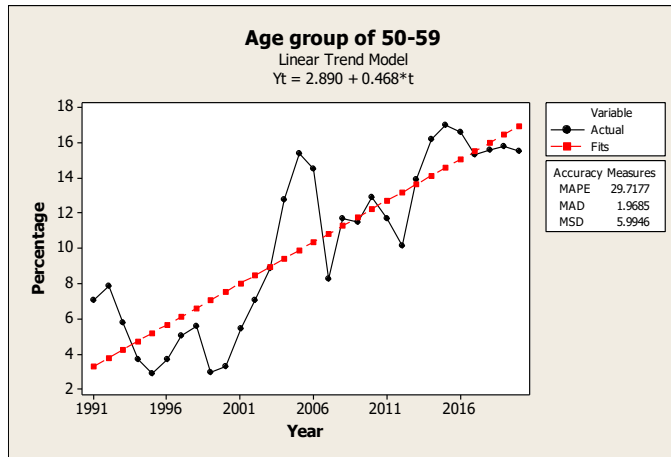


Table 5: Regression Parameters for Annual Percentage Distribution of Tourists by Age Groups

Age Groups	m (Slope)	c (Intercept)	R-square (R ²)	Trend	Strength of relationship
3 - 19	0.0406	8.26	0.028	Increase	Very weak
20 – 29	- 0.787	36.6	0.712	Decrease	Very strong
30 - 39	- 0.281	31.6	0.488	Decrease	Moderate
40 - 49	0.336	13.8	0.265	Increase	Weak
50 - 59	0.468	2.89	0.733	Increase	Very strong
60 & over	0.221	6.88	0.288	Increase	Weak

Occupation

Figure 7 shows the trends in the percentage distribution of tourists by Occupational Categories between 1991 to 2019. The highest rate of decline in the annual percentage of tourist arrivals by Occupational Categories is demonstrated by tourists in the "Other Occupation", with a rate of -0.727 percent per year. It is shown that the trend of tourists who are in other occupation has had a strong decline. Table 6 shows that the highest rate of increasing trend in annual percentage is shown in the occupational categories "professionals" category, with a rate of 0.942 percent per year while the lowest rate of increasing trend in the "Retired persons" category, with a rate of 0.0132 percent per year, which is only a slight increase as can be noted from Figure 7. The trend analyses show that all the occupational categories, except Businessmen, No Occupation, and Other Occupation categories have an increasing trend in the annual percentage distribution of the tourists who arrived in Sri Lanka during the period from 1991 to 2019 (Table 1).

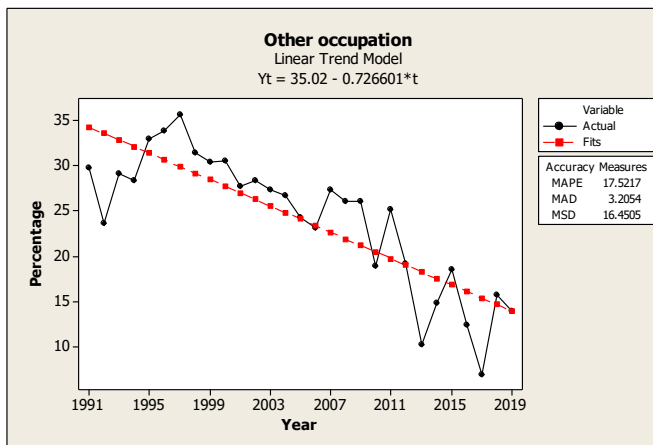
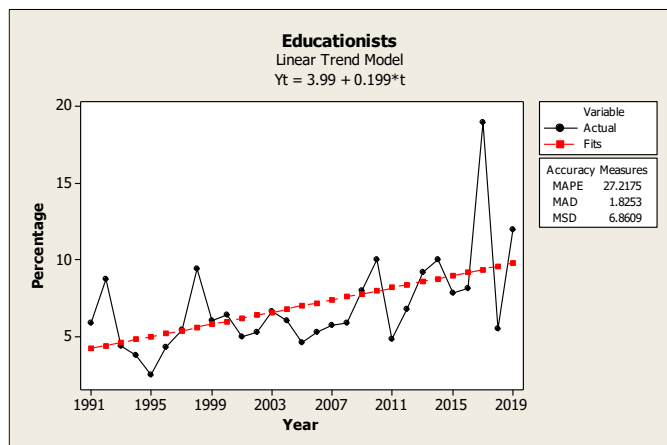
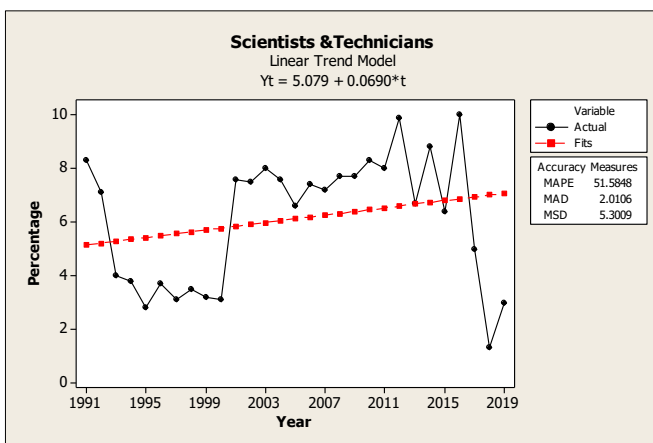
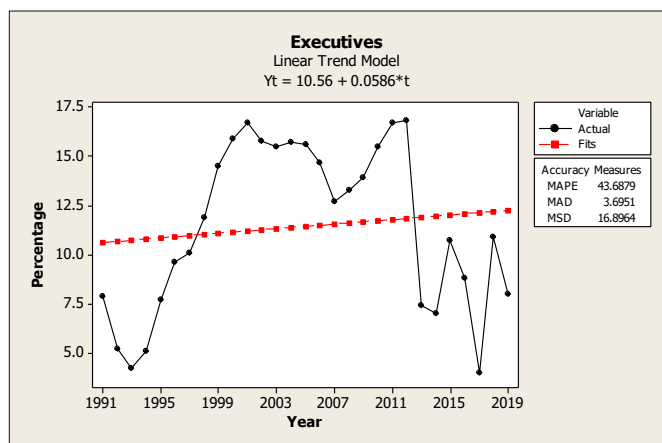
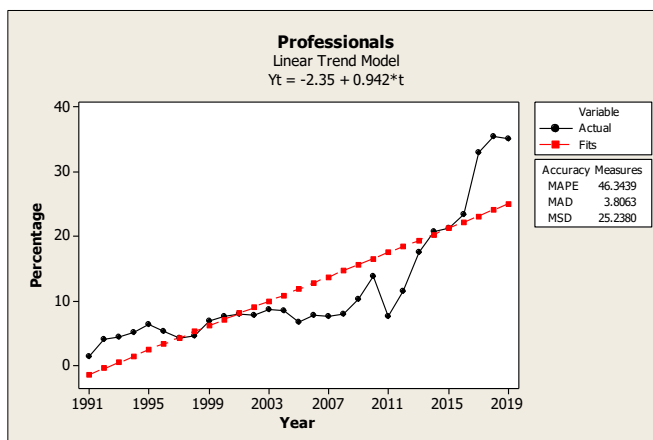
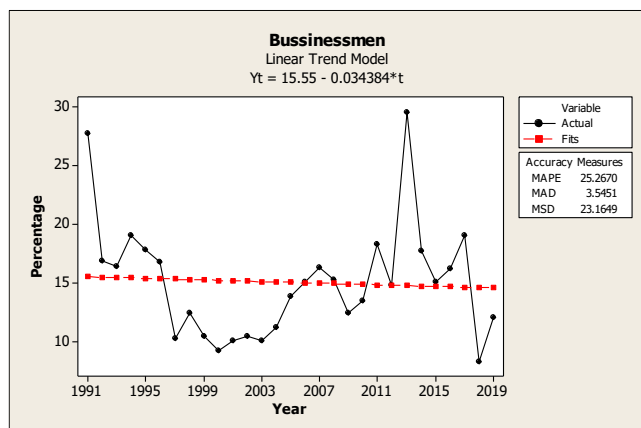
The categories of “No occupation” and “Other occupation” have a strong decrease in the annual percentage of tourist arrivals with R^2 values are 0.677 and 0.692 respectively while “Businessmen” have shown a low decline during the study period. It is also shown that the occupational category of “Professionals” has had a very strong positive trend with a rate of 0.942 percent per year. However, the occupational category of “Educationists,” “Scientists & Technicians,” “Executives” and “Retired persons” have a slightly increasing trend. It is displayed in Figure 7. According to the facts revealed from the statistical analysis, it is noticeable that the majority of the tourists who arrived in Sri Lanka during the period were gainfully occupied which has shown a significant increasing trend in the annual percentage distribution of tourists.

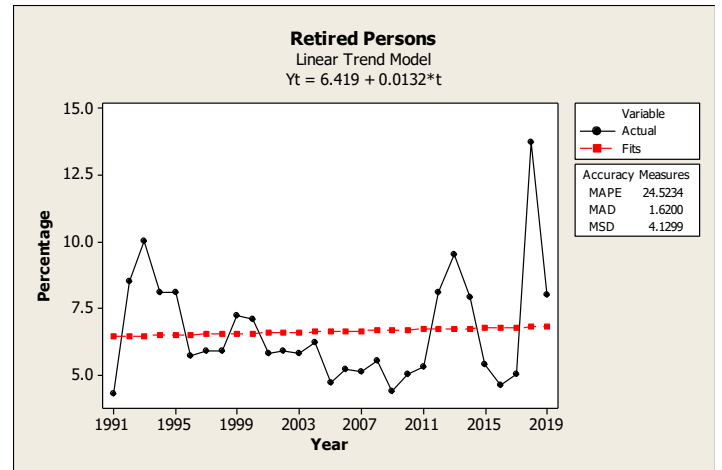
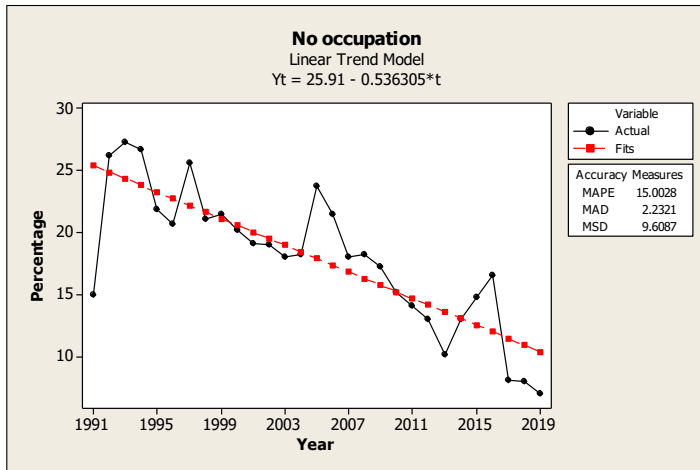
Table 6: *Regression Parameters for Annual Percentage Distribution of Tourists by Occupational Categories*

Occupation	m (Slope)	c (Intercept)	R-square (R^2)	Trend	Strength of relationship
Professionals	0.942	-2.35	0.711	Increase	Very Strong
Educationists	0.199	3.99	0.289	Increase	Weak
Scientists & Technicians	0.0690	5.08	0.059	Increase	Very Weak
Executives	0.0586	10.6	0.014	Increase	Very Weak
Retired persons	0.0132	6.42	0.003	Increase	Very Weak
Businessmen	-0.034	15.5	0.004	Decrease	Very Weak
No Occupation	-0.536	25.9	0.677	Decrease	Strong
Other Occupation	-0.727	35.0	0.692	Decrease	Strong

Note: Data have been arranged in descending order based on the slope

Figure 7: Trends in Percentage Distribution of Tourists by Occupational Categories – 1991 to 2019



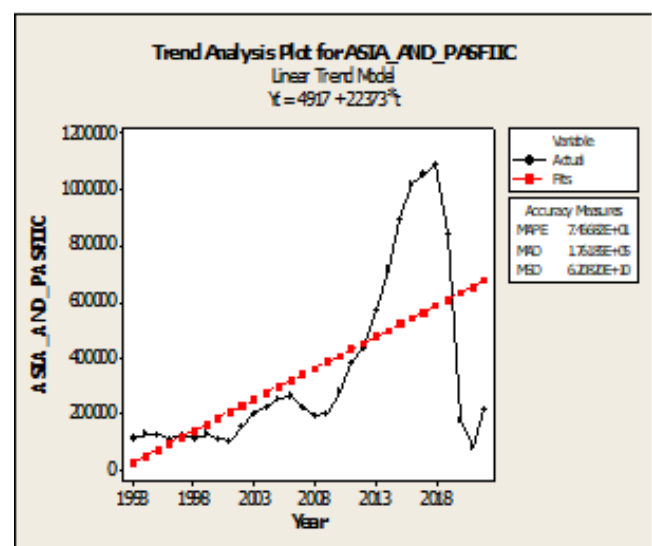
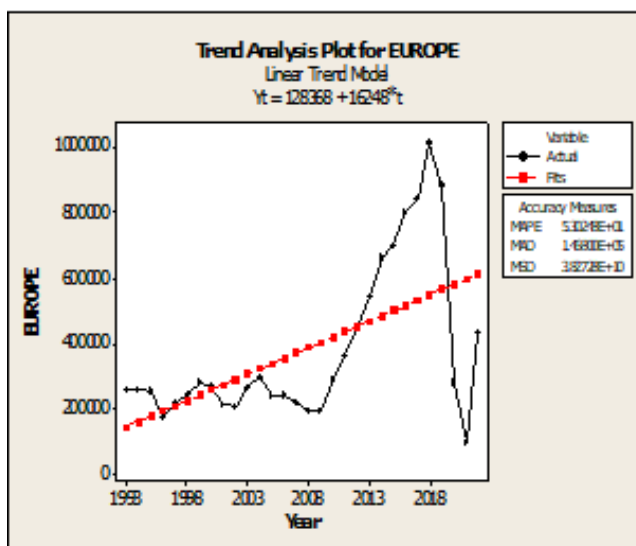


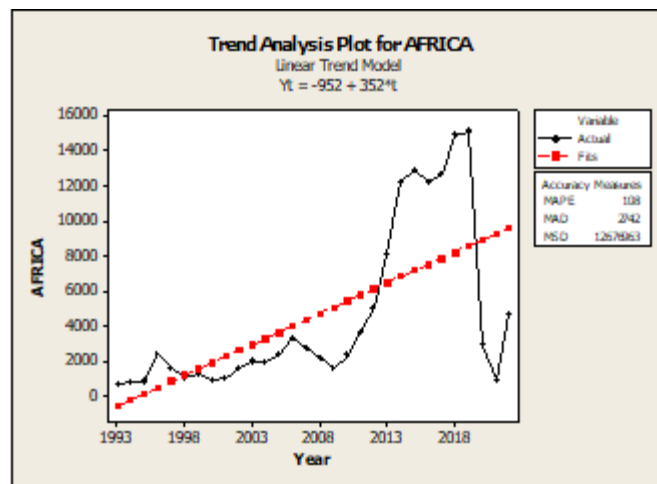
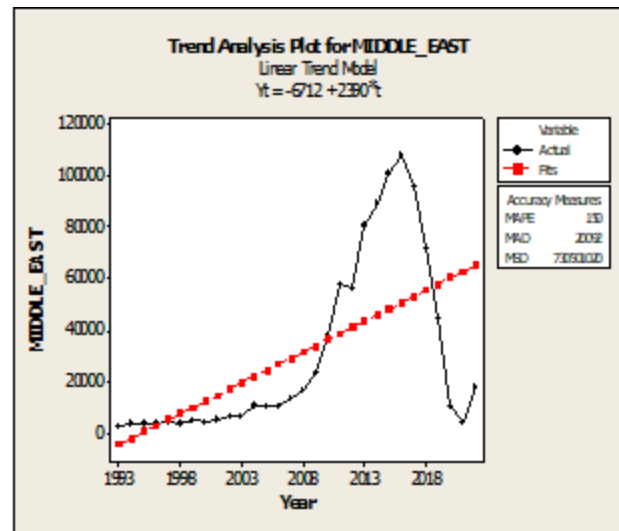
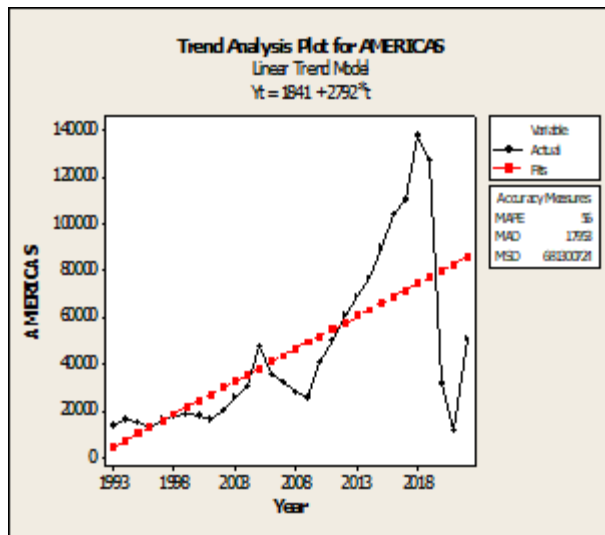
Different Regions

Table 7: Regression Parameters for Annual Tourist Arrivals by Different Regions

Region	m(Slope)	c(Intercept)	R-Square(R^2)	Trend	Strength of Relationship
AMERICAS	2792	1841	0.462	Increase	Moderate
EUROPE	16248	128368	0.341	Increase	Moderate
AFRICA	352	-952	0.422	Increase	Moderate
MIDDLE_EAST	2390	2390	0.369	Increase	Moderate
ASIA and PACIFIC	22373	4917	0.377	Increase	Moderate

Figure 8: Trends in Tourist Arrivals by Regions – 1993 to 2023





When considering the regional distribution of tourist arrivals, it is observable that all regions have shown a moderate increasing trend in the annual number of tourist arrivals (Table 7). Specifically, from 1993 to 2023, all regions have demonstrated a moderate increase. Tourists from the Asia and Pacific region have shown the highest annual growth, with an increase of 22,373 arrivals per year. One possible reason for this growth is the peaceful environment in Sri Lanka following the end of the civil war.

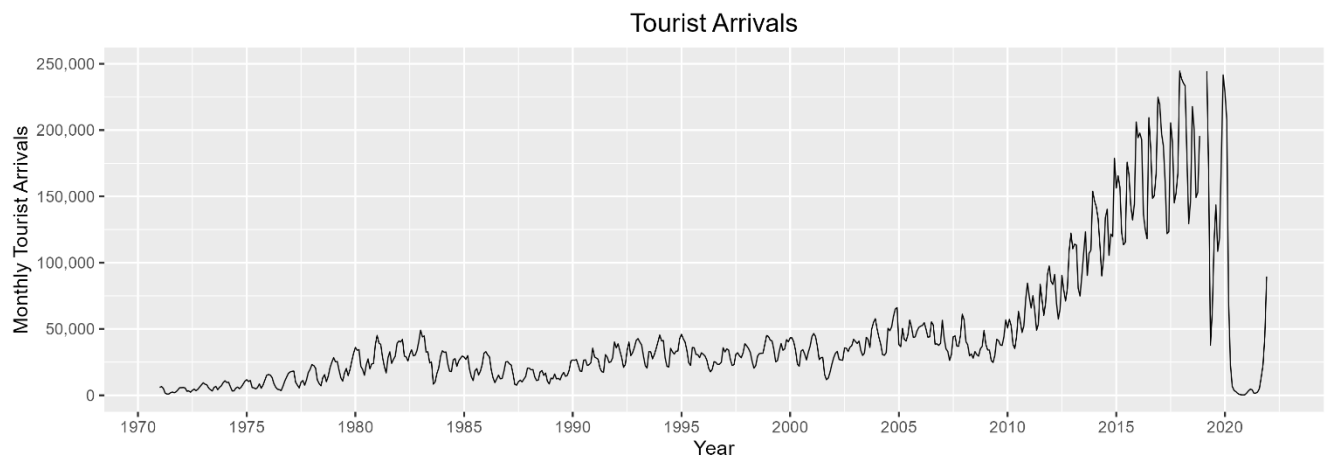
Following the Asia and Pacific region, tourists from Europe and the Americas have also shown significant growth in their arrivals, with rates of 16,248 and 2,792 arrivals per year, respectively. These trends are also considered moderate.

However, during the final years of the civil war from 2006 to 2009, tourist arrivals from all regions except the Middle East decreased. This decline was because countries in these regions considered Sri Lanka an unsafe destination and prohibited their tourists from visiting during that time. Among the regions, Africa recorded the lowest number of arrivals with an increasing trend, at a rate of 352 arrivals per year. The results also reveal a significant increase in tourist arrivals from all regions after the year 2009. However, tourist arrivals from all regions declined between 2020 and 2022 due to the COVID-19 pandemic, which affected the entire world.

Data and Forecast Overview

Data from January 1971 to December 2021 is used as the model estimation sample, while the most recent data from January 2022 to December 2023 is set aside for the evaluation and validation of the forecasting model. The objective is to develop the best model to forecast tourist arrivals and Figure 9 shows monthly tourist arrivals from 1971 to 2021. Tourist arrivals are significantly lower in 2020. Sri Lanka has never seen such a significant drop in tourist arrivals from its usual trend. The next step is to select a model and forecast the monthly tourist data. Due to the COVID-19 pandemic, airports were closed from April 2020 to December 2020, resulting in a lack of data on tourist arrivals during this period. Consequently, there are missing values in the dataset for these months. We addressed this issue by estimating the missing values using R software, and these estimated values were subsequently used in the modeling process.

Figure 9: *monthly tourist arrivals 1971-2021*



Model Selection:

We tested the stationarity of the original dataset using the Augmented Dickey-Fuller (ADF) test. The maximum lag order for the regression model was set to 20 lags. The ADF test returned a p-value of 0.131. Since this p-value is greater than the significance level ($\alpha = 0.05$), we did not reject the null hypothesis, indicating that tourist arrivals from January 1971 to December 2021 are non-stationary.

Augmented Dickey-Fuller Test

Null hypothesis: Data are non-stationary

Alternative hypothesis: Data are stationary

Test

Statistic	P-Value	Recommendation
-2.43973	0.131	Test statistic > critical value of -2.86644. Significance level = 0.05 Fail to reject the null hypothesis. Consider differencing to make data stationary.

In this step, we performed a first non-seasonal differencing to remove the trend, followed by another ADF test. The ADF test indicates that the p-value is 0.000, which is less than the significance level ($\alpha = 0.05$). As a result, we rejected the null hypothesis and concluded that the dataset is stationary.

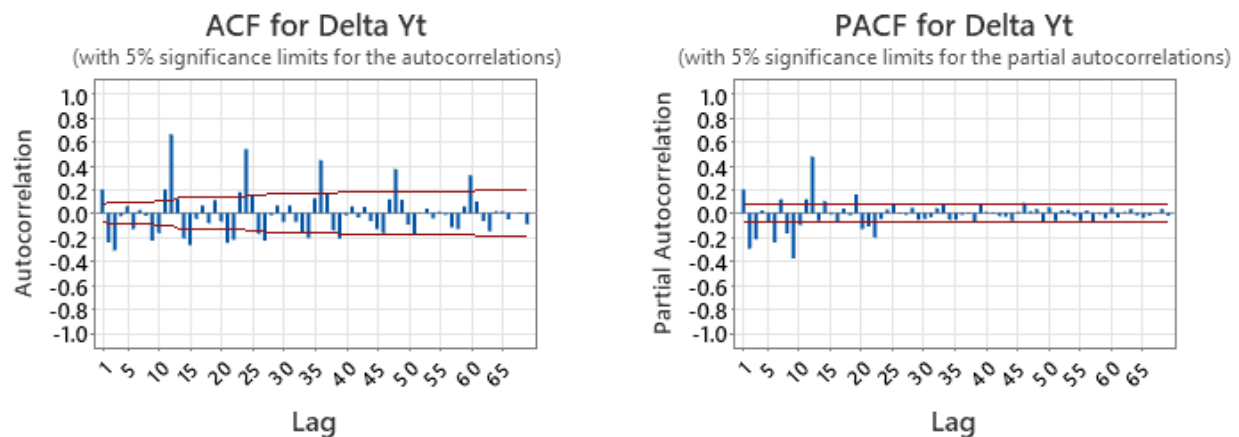
Augmented Dickey-Fuller Test

Null hypothesis: Data are non-stationary
 Alternative hypothesis: Data are stationary

Test		
Statistic	P-Value	Recommendation
-4.96779	0.000	Test statistic $\leq$ critical value of -2.86645. Significance level = 0.05 Reject null hypothesis. Data appears to be stationary, not supporting differencing.

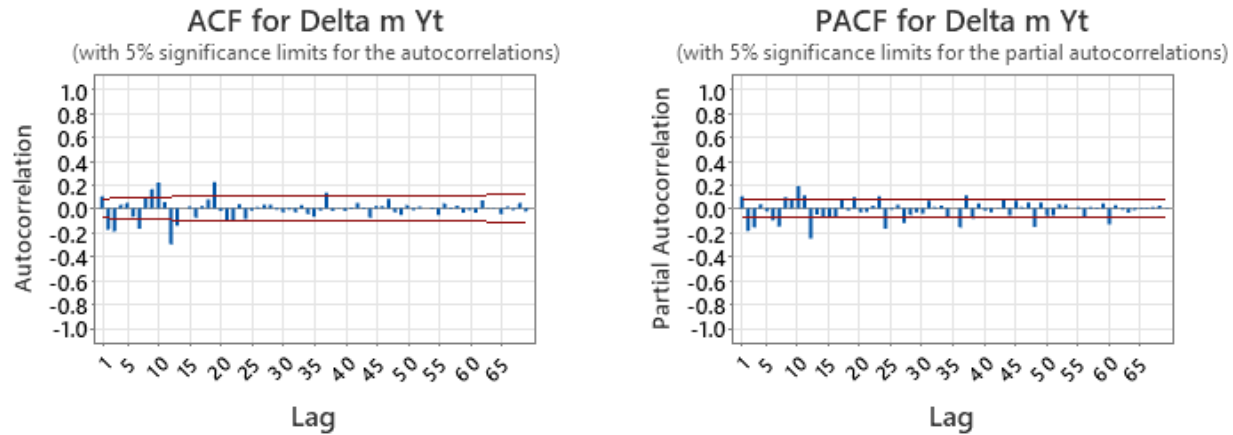
Then, we analyzed the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots. The ACF plot (Figure 10) indicated a seasonality with a cycle length of 12 months. To remove this seasonality, we applied a first seasonal differencing.

Figure 10: ACF and PACF plots after applying non-seasonal lag 1 differencing



The AR component was determined from the Partial Correlation, while the MA (Moving Average) component was derived from the Autocorrelation. The first seasonal lag in the ACF plot was negative, leading us to assign SAR (0). We then examined the number of significant seasonal lags (12, 24, 36, etc.), and since the number of significant seasonal lags was 1, we assigned SMA (1).

Figure 11: *ACF and PACF plots after applying both non-seasonal and seasonal lag 1 differencing*



This study suggests several models, and the most fitted model is chosen for the analysis described in the estimation section.

Estimation:

This analysis suggested several models, from which the best model was selected based on the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) values. The selected models include:

- $ARIMA(0,1,0)(0,1,1)_{12}$
- $ARIMA(0,1,0)(0,1,2)_{12}$
- $ARIMA(3,1,0)(1,1,2)_{12}$
- $ARIMA(3,1,0)(0,1,2)_{12}$

Table 8: *Summary of the selected SARIMA models*

Model	AIC	BIC
$ARIMA(0,1,0)(0,1,1)_{12}$	12956.84	12965.63
$ARIMA(0,1,0)(0,1,2)_{12}$	12937.56	12950.75
$ARIMA(3,1,0)(1,1,2)_{12}$	12892.98	12923.75
$ARIMA(3,1,0)(0,1,2)_{12}$	12891.07	12917.44

ARIMA(0,1,0)(0,1,1)₁₂ :

Type	Coef	SE Coef	T-Value	P-Value
SMA 12	0.6598	0.0328	20.14	0.000
Constant	-41	170	-0.24	0.811

ARIMA(0,1,0)(0,1,2)₁₂ :

Type	Coef	SE Coef	T-Value	P-Value
SMA 12	0.5273	0.0436	12.09	0.000
SMA 24	0.2339	0.0473	4.95	0.000
Constant	-7	120	-0.06	0.951

ARIMA(3,1,0)(1,1,2)₁₂ :

Type	Coef	SE Coef	T-Value	P-Value
AR 1	0.0945	0.0407	2.32	0.021
AR 2	-0.1800	0.0406	-4.44	0.000
AR 3	-0.1907	0.0406	-4.69	0.000
SAR 12	0.115	0.246	0.47	0.641
SMA 12	0.633	0.249	2.54	0.011
SMA 24	0.193	0.185	1.04	0.298
Constant	-6.7	83.7	-0.08	0.937

ARIMA(3,1,0)(0,1,2)₁₂ :

Type	Coef	SE Coef	T-Value	P-Value
AR 1	0.0927	0.0406	2.28	0.023
AR 2	-0.1804	0.0404	-4.47	0.000
AR 3	-0.1921	0.0405	-4.74	0.000
SMA 12	0.5196	0.0440	11.82	0.000
SMA 24	0.2740	0.0481	5.69	0.000
Constant	-9.3	99.5	-0.09	0.926

Finally, we compared the actual international tourist arrivals with the forecasted values from January 2022 to December 2023. A table has been provided showing the Mean Absolute Percentage Error (MAPE) and Mean Absolute Error (MAE) values for each model. Therefore, the selected ARIMA(0,1,0)(0,1,2)₁₂ model is appropriate and it perform well in predicting international tourist arrivals in Sri Lanka.

Table 9: Forecast error comparison of the competing models

Model	MAE	MAPE
ARIMA(0,1,0)(0,1,1) ₁₂	76716.16	85.83715
ARIMA(0,1,0)(0,1,2) ₁₂	32377.18	34.36147
ARIMA(3,1,0)(1,1,2) ₁₂	32348.65	34.13186
ARIMA(3,1,0)(0,1,2) ₁₂	33105.91	35.07289

Figure 12: Actuals vs Forecasts of the selected model (training set)

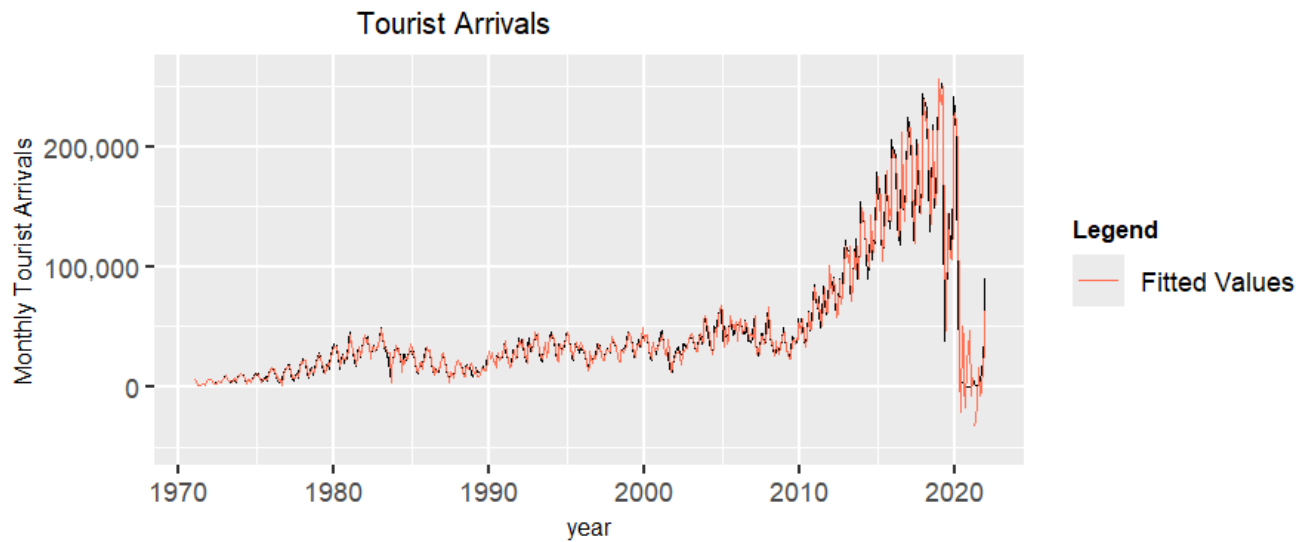
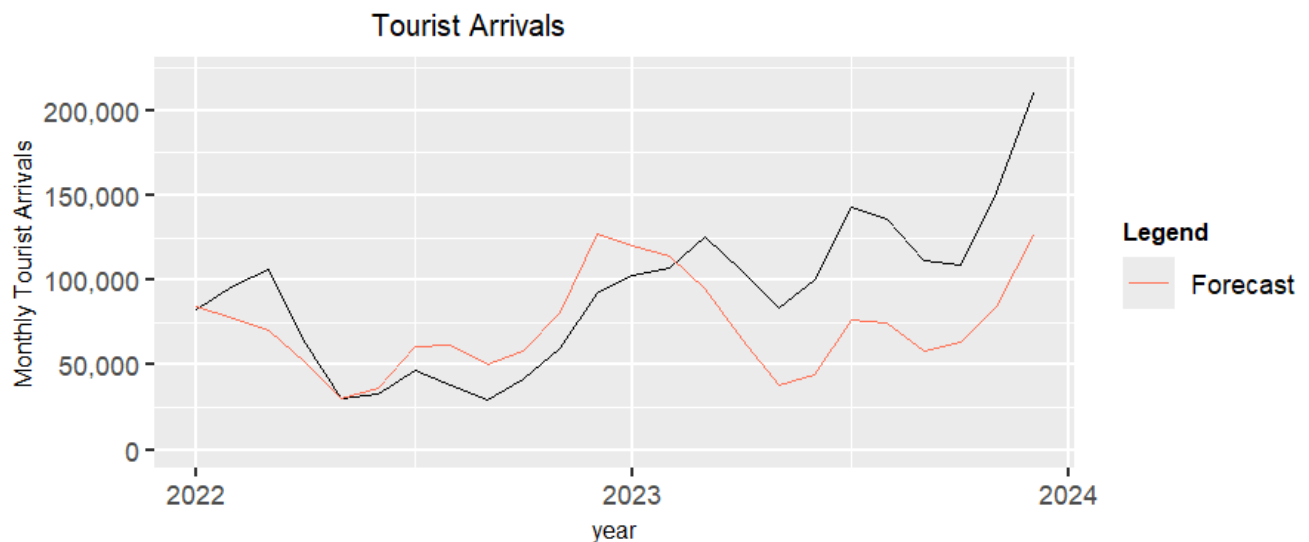


Figure 13: Actuals vs Forecasts of the selected model (testing test)



5. Discussion

The analysis of tourist arrivals in Sri Lanka from 1971 to 2023 reveals several key insights, influenced by significant external events, trend analysis methods, and model selection criteria.

Impact of the COVID-19 Pandemic and the 2019 Easter Bombings

The COVID-19 pandemic in 2020 and the Easter bombings in April 2019 had profound impacts on Sri Lanka's tourism industry. The Easter bombings resulted in an immediate and sharp decline in tourist arrivals due to concerns over safety. Just as the industry was beginning to recover, the onset of the COVID-19 pandemic in early 2020 led to unprecedented global travel restrictions, border closures, and lockdowns, further exacerbating the decline in tourist arrivals. During the period from April 2020 to December 2020, there was a significant lack of data on tourist arrivals, as the country faced strict lockdowns and travel bans. These missing values presented a challenge in the analysis. To address this, estimations were made for this period, allowing for a more continuous analysis despite the data gaps. However, it's important to acknowledge that these estimates may introduce a level of uncertainty into the analysis, given the unique and extreme nature of the circumstances during this time.

Trend Analysis

In conducting the trend analysis, only linear trends were considered. The linear trend analysis provided a straightforward understanding of the general direction of tourist arrivals over the years. The results indicated a moderate increase in tourist arrivals, with the number growing continuously over time. However, the decision to limit the analysis to linear trends means that the potential presence of more complex, higher-order trends was not explored. This choice may have overlooked cyclical patterns, non-linear growth, or other nuanced trends that could have provided additional insights into the dynamics of tourist arrivals. The limitation to linear trends simplifies the interpretation but at the cost of potentially missing more intricate patterns in the data.

SARIMA Model Selection

For forecasting purposes, a Seasonal Autoregressive Integrated Moving Average (SARIMA) model was selected based on the values of BIC (Bayesian Information Criterion), AIC (Akaike Information Criterion), MAPE (Mean Absolute Percentage Error), and MAE (Mean Absolute Error). The use of these criteria ensures that the model chosen is not only statistically strong but also minimizes errors in forecasting. However, the selection process focused totally on these statistical measures, without considering other potential factors such as external variables like the pandemic or the Easter bombings. Although BIC, AIC, MAPE, and MAE are useful for choosing a model, focusing only on these criteria might cause the model to miss some important details, especially those affected by unexpected events. Although BIC, AIC, MAPE, and MAE are useful for choosing a model, focusing only on these criteria might cause the model to miss some important details, especially those affected by unexpected events.

6. Conclusions

6.1 Summary of Findings

This comprehensive analysis provides insights into the trends and changes in Sri Lanka's tourism industry over several decades, illustrating shifts in visitor numbers, seasonal patterns, purposes of visits, duration of stays, demographic trends, and regional origins of tourists. The analysis of annual tourist arrivals in Sri Lanka from 1971 to 2023 indicates a moderate upward trend, with an average yearly increase of approximately 26,452 tourists. The analysis of tourist arrivals in Sri Lanka from 1971 to 2023 reveals clear seasonal patterns. The 12-month moving average method shows that tourist arrivals peak in December and January, with seasonal indices of 1.306 and 1.263, respectively. Tourist numbers are lowest in May and June, with indices around 0.7. A gradual rise starting in October, culminating in a December peak while there is a gradual fall from March to May.

The analysis reveals a decrease in tourists visiting for pleasure/holiday and religious/cultural purposes, while there is an increase in those visiting for business and to see friends and relatives. The trends indicate changing motivations for travel to Sri Lanka over time. Tourists staying for shorter durations (1-3 nights and 4-7 nights) show an increasing trend, while longer stays (15-21 nights, 22-30 nights, and over 31 nights) show a decreasing trend. This shift suggests a trend towards shorter visits. Male tourist representation has been moderately decreasing, while female representation has been moderately increasing. Younger tourists (ages 20-39) show a declining trend, while older age groups (50-59 and 60+) show an increasing trend, indicating a shift towards an older tourist demographic. Tourists classified as "Professionals" show a strong increasing trend, while those with "No Occupation" or "Other Occupations" show a significant decline. The analysis highlights a growing proportion of professionally employed tourists visiting Sri Lanka. Tourist arrivals from all regions have shown moderate growth, with the Asia and Pacific region leading the increase. The data reflects significant recovery and growth post-civil war (2009) despite a temporary decline during the COVID-19 pandemic.

This study aimed to develop the best model for forecasting monthly tourist arrivals in Sri Lanka. Data from January 1971 to December 2021 was used for model estimation, while data from January 2022 to December 2023 was set aside for validation. The analysis began by testing the stationarity of the data using the Augmented Dickey-Fuller (ADF) test, which indicated that the data was non-stationary. After applying first non-seasonal and then seasonal differencing, the data became stationary. Several ARIMA models were tested, and the best model was selected based on the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). The selected model, ARIMA(0,1,0)(0,1,2)₁₂, performed well in predicting tourist arrivals, showing lower Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) compared to other models when validated against data from 2022 to 2023. Based on our forecast, visitor arrivals in Sri Lanka will gradually increase over the next years.

6.2 Recommendations for Stakeholders

For the government and tourism authorities, it's important to improve marketing by encouraging tourists to visit during less busy times of the year. This can help even out the flow of tourists throughout the year. Developing new types of tourism, like eco-tourism, wellness tourism, and adventure tourism, can also attract different kinds of visitors. Investing in better infrastructure, such as transportation, accommodation, and tourist facilities, will make the experience better for visitors and help handle the growing number of tourists. It's also crucial to promote tourism that is environmentally friendly and respectful of local cultures, to protect Sri Lanka's natural beauty and heritage.

For tourism businesses, it's recommended to offer tailored experiences that meet the interests of different groups. For example, younger tourists might enjoy adventure activities, while older visitors might prefer cultural experiences. Businesses should also consider expanding their services beyond just leisure tourism. Offering options like business tourism or event-based tourism can attract more visitors and provide more stability throughout the year.

For local communities, it's a good idea to get involved in tourism by offering unique local experiences, products, and services that showcase Sri Lankan culture. This not only makes the visit more enjoyable for tourists but also ensures that local communities benefit from the tourism industry.

6.3 Areas for Future Research

While this study offers valuable insights into tourist trends in Sri Lanka, several areas merit further exploration. One important area is the impact of external factors on tourist arrivals. Future research could focus on how global events, such as pandemics or economic crises, affect tourism and what strategies can help the industry build resilience. Additionally, understanding tourist satisfaction and behavior is crucial. Studying the factors that influence these aspects can help tailor tourism products and services more effectively, leading to improved experiences for visitors. By understanding these trends, stakeholders can better position Sri Lanka as an attractive destination for these emerging groups, ensuring sustained development in the tourism industry.

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